USER REQUIREMENT SPECIFICATION

Indian Energy Monitoring System

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ABSTRACT

The Indian Energy Monitoring System is a comprehensive web-based platform designed to track, analyze, and visualize India's energy generation landscape in real-time. The system aggregates live data from meritindia.in, capturing daily generation statistics from renewable and conventional power plants across multiple states and regions. Built with a React-based frontend and Flask backend architecture, the platform provides stakeholders with interactive dashboards, analytical insights, and automated alert mechanisms for monitoring energy production patterns. The system maintains a structured database of power plants, companies, and regional metrics, enabling comparative analysis of energy mix, generation trends, and capacity utilization. By processing daily XLS and PDF reports through automated web scraping, the platform transforms raw energy data into actionable intelligence, supporting decision-making for energy policy, grid management, and renewable energy transition planning. The application serves as a centralized monitoring tool for understanding India's dynamic energy ecosystem and facilitating data-driven insights into the nation's power generation infrastructure.

1. SYSTEM OVERVIEW

A comprehensive real-time energy monitoring platform that tracks, analyzes, and visualizes India's power generation infrastructure by aggregating data from meritindia.in, providing stakeholders with actionable insights for energy management and policy planning.

2. FUNCTIONAL REQUIREMENTS

2.1 Data Acquisition & Processing

- **Real-time Data Scraping:** Automated web scraping from meritindia.in to fetch daily power generation reports
- **Multi-format Support:** Process XLS (plant generation) and PDF (renewable energy) reports
- **Scheduled Updates:** Daily synchronization of power plant generation data from GitHub repository
- **Data Parsing Pipeline:** Sequential execution of integrated_web_scrapping.py, parseall1.py, parseall2.py, and parseall3.py

2.2 Dashboard & Visualization

Main Dashboard KPIs:

- Total number of power plants
- Today's generation (MU - Million Units)
- Today's demand (MU)
- Average efficiency percentage
- Total operational capacity (MW - Megawatts)
- Plants under outage count
- Critical coal stock alerts

- Date-based filtering capability

Interactive Charts:

- Energy mix pie chart (renewable vs non-renewable)
- Regional performance bar charts
- Trend line charts for historical analysis
- Map visualization for geographical distribution
- Top performing plants comparison

2.3 Plant Management

Plant Listing:

- Paginated view (20 plants per page)
- Search functionality by plant name/ID
- Filter by state, sector (State/Private/Central), energy type
- Display basic plant information

Detailed Plant View:

- Comprehensive plant information (name, location, sector, energy type)
- Production history with efficiency metrics
- Operational capacity and capability
- Coal stock levels (for thermal plants)
- Operational status by unit
- Outage information with expected sync dates
- Date-based historical data viewing

2.4 Analytics & Insights

Regional Performance Analysis:

- Region-wise total plants count
- Generated energy (MU)
- Demand (MU)
- Surplus/deficit calculation
- Supply percentage metrics
- Energy status classification

Efficiency Comparison:

- Plant-wise average efficiency ranking
- Top 20 performing plants
- Efficiency percentage over time

Renewable Energy Mix:

- State-wise renewable vs non-renewable breakdown
- Renewable percentage calculation
- Production values in MU
- Last 30-day aggregated data

2.5 Alert System

Real-time Monitoring Alerts:

- **Low Efficiency Alerts:** Plants with <60% average efficiency (3-day window)
- **Critical Coal Stock:** Thermal plants with <7 days coal stock (WARNING: 4-7 days, CRITICAL: <4 days)
- **Plant Outage Alerts:** Units under outage status (CRITICAL: ≥500 MW, WARNING: <500 MW)

Alert Dashboard:

- Total alerts count
- Critical vs warning categorization
- Detailed alert information with plant name, state, and values
- Coal stock table with operational capacity

2.6 Regional/State Management

- Plant count per state
- Generated energy

- Demand
- Energy status (surplus/deficit)
- Date-based filtering
- Regional grouping (Northern, Southern, Eastern, Western, North-Eastern)

2.7 Database Administration

- **Daily Report Generation:** Generate comprehensive daily energy reports by date with state-wise aggregated metrics
- **Underperforming Plant Identification:** Configurable efficiency threshold and time window
- **Data Pipeline Trigger:** Manual trigger for complete ETL process with error handling

3. DATA MODEL

3.1 Core Entities

- **POWERPLANTS:** Plant_ID, Plant_Name, State_Code, Sector_ID, Type_ID
- **STATE:** State_Code, State_Name, Region, Population
- **SECTOR:** Sector_ID, Sector_Name (State/Private/Central)
- **ENERGYTYPE:** Type_ID, Type_Name (Thermal, Hydro, Wind, Solar, Biomass, etc.), Description
- **PRODUCTIONLOG:** Plant_ID, Log_Date, Efficiency_Percentage, Todays_Actual_MU, Capable_Generation_MU, Operational_Capacity_MW, Coal_Stock_Days
- **OPERATIONAL_STATUS:** Plant_ID, Unit_Number, Status_Date, Cap_Under_Outage_MW, Status, Outage_Date, Expected_Sync_Date, Remarks
- **REGION_DETAILS:** State_Code, Report_Date, Generated_MU, Imported_MU, Surplus_MU, Demand_MU, Monitored_Capacity_MW, Grid_Frequency_HZ
- **DATE_DIM:** Date, Day, Month, Year

4. TECHNICAL ARCHITECTURE

4.1 Backend (Python/Flask)

- **Framework:** Flask 3.0.0
- **Database:** MySQL with SQLAlchemy ORM
- **API Architecture:** RESTful APIs with Blueprint modularization
- **Routes:** /api/dashboard/*, /api/plants/*, /api/alerts/*, /api/analytics/*, /api/regions/*, /api/db-admin/*, /api/health

4.2 Frontend (React)

- **Framework:** React 18.3.1
- **Routing:** React Router DOM 7.9.5
- **Charts:** Recharts 3.3.0
- **HTTP Client:** Axios 1.13.2
- **UI Components:** Reusable Card components, Loading spinners, Alert banners, Data tables with pagination, Interactive charts, Navbar and Sidebar navigation

4.3 Data Processing

- **Web Scraping:** Requests, PDFPlumber, Pandas
- **Data Sources:** GitHub repository for daily XLS reports, meritindia.in for renewable PDF reports
- **File Formats:** XLS, PDF, CSV
- **Output Folders:** Daily_Plant_Generation_XLS_Reports/, Daily_Renewable_PDF_Reports/, Processed_Renewable_XLSX_reports/

5. USER INTERFACES

5.1 Pages

- **Dashboard** - Main overview with KPIs and charts
- **Plants** - Searchable, filterable plant listing
- **PlantDetails** - Detailed single plant view
- **Analytics** - Regional performance and renewable mix analysis
- **Alerts** - System alerts and critical warnings
- **Regions** - State-wise energy statistics
- **Companies** - Company/sector information
- **Admin** - Database administration tools

5.2 Navigation

- Fixed sidebar with page links
- Top navbar with branding
- Responsive design for desktop/tablet

6. NON-FUNCTIONAL REQUIREMENTS

6.1 Performance

- API response time <2 seconds for dashboard queries
- Support for pagination to handle large datasets
- Efficient database queries with proper indexing

6.2 Reliability

- Error handling for failed web scraping
- Graceful degradation when data is unavailable
- Retry mechanisms for API failures
- Data validation at input level

6.3 Usability

- Intuitive dashboard with visual KPIs
- Date picker for historical data viewing
- Search and filter capabilities
- Loading indicators during data fetch
- Error messages with retry options

6.4 Maintainability

- Modular code structure
- Separation of concerns (routes, services, models)
- Reusable React components
- Environment-based configuration
- Comprehensive error logging

6.5 Security

- CORS configuration for API access
- SQL injection prevention via parameterized queries
- Input validation and sanitization

7. SYSTEM CONSTRAINTS

- MySQL database required
- Daily data updates from meritindia.in
- Internet connectivity for web scraping
- Storage for XLS/PDF report archives
- Python 3.x runtime environment
- Node.js for frontend development